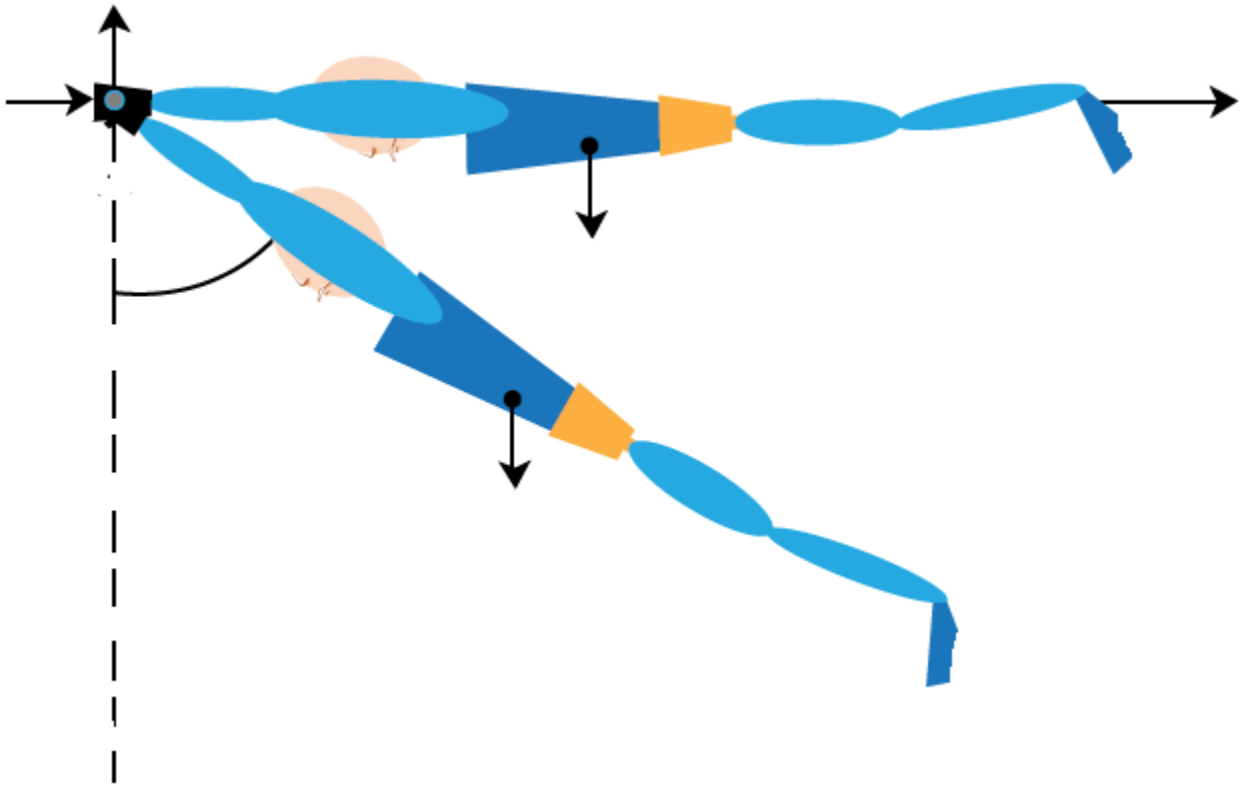
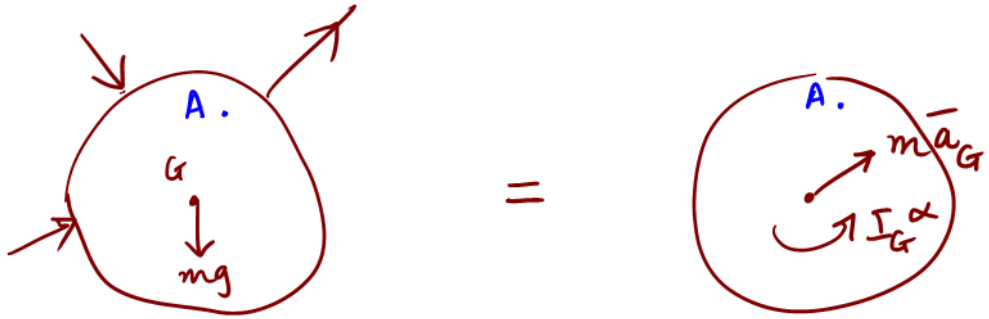
Angular motion – distributed mass, two-rod model





$\sum M_A = (\sum M_A)_K$

$= I_G\bar{\alpha} + \bar{r}_{A/G} \times m\bar{a}_G$

This will reduce to

$\sum M_O = I_O\alpha$ if O is a fixed point

& $\sum M_G = I_G\alpha$ if G is the CoM

References

INTRODUCTION

- *Basic Biomechanics* by Susan J. Hall
- *Biomechanical Basis of Human Movement* by Joseph Hamill and Kathleen Knutzen

STATICS

- *Fundamentals of Biomechanics: Equilibrium, Motion, and Deformation* by Nihat Ozkaya and Margareta Nordin
- *Human Factors Engineering* by Chandler Allen Phillips

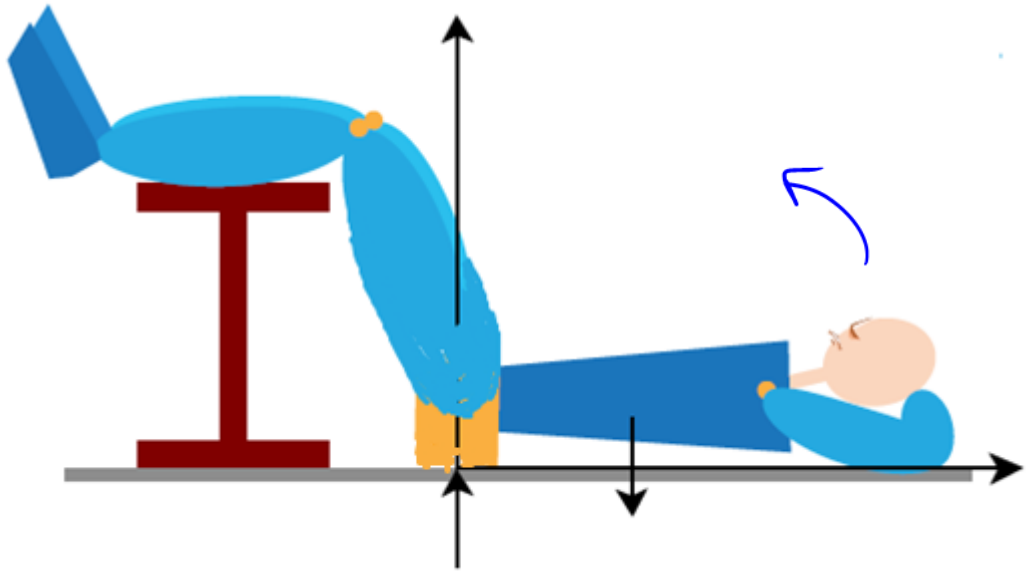
DYNAMICS

- *Human Body Dynamics* by Aydin Tozeren
- *Biomechanical Analysis of Fundamental Human Movements* by Arthur E. Chapman

HUMAN WALKING

- *Gait Analysis* by Michael Whittle
- *Gait Analysis: Normal and Pathological Function* by Jacquelin Perry and Judith M. Burnfield

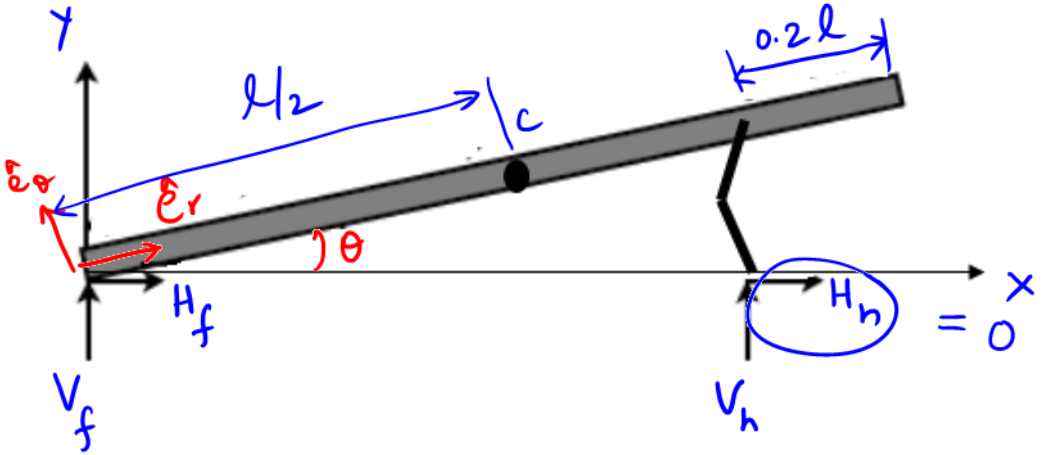
Sit-ups



Push-ups

Person modeled as a slender uniform rod of length l & mass m. Arms modeled as two weightless rods. Find the reaction forces at the hands and feet





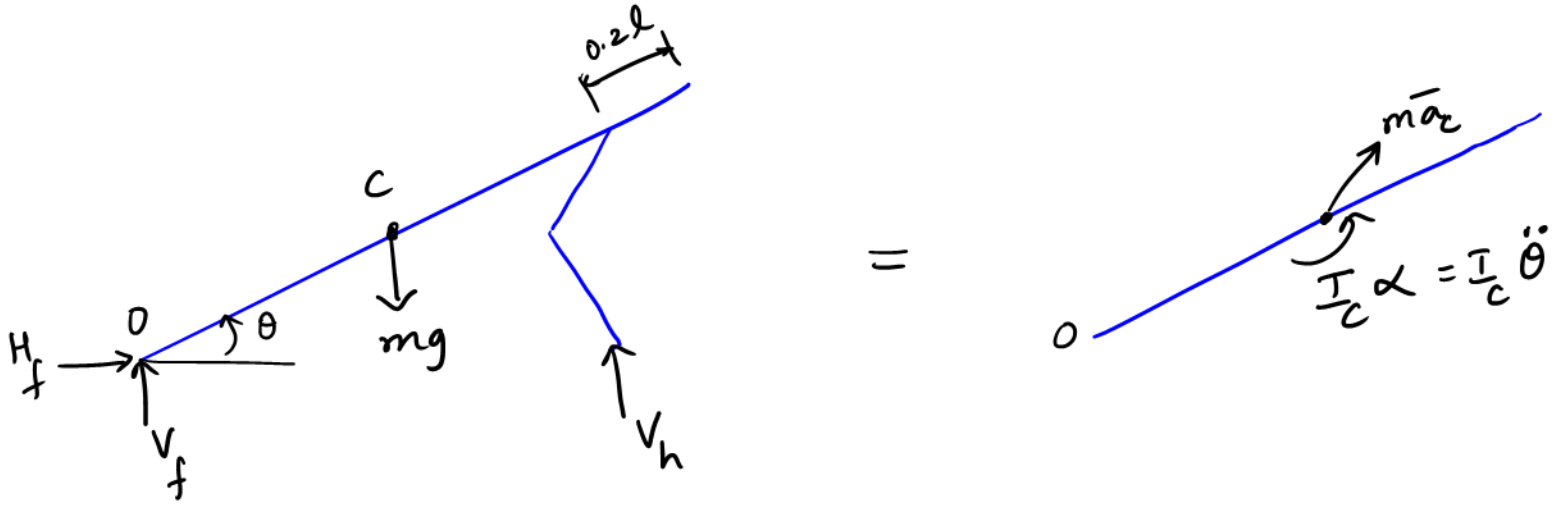
$$\hat{e}_r = \cos\theta\,\hat{i} + \sin\theta\,\hat{j} \qquad \dot{\hat{e}}_r = \dot{\theta}\,\hat{e}_\theta$$

$$\hat{e}_\theta = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j} \qquad \dot{\hat{e}}_\theta = -\dot{\theta}\,\hat{e}_r$$

$$\bar{r}_c = \frac{l}{2}\,\hat{e}_r$$

$$\bar{v}_c = \frac{l}{2}\,\dot{\theta}\,\hat{e}_\theta$$

$$\bar{a}_c = \frac{l}{2}\,\ddot{\theta}\,\hat{e}_\theta - \frac{l}{2}\,\dot{\theta}^2\,\hat{e}_r$$



$$\sum \bar{M}_o = I_C \ddot{\theta}\,\hat{k} + \bar{r}_c \times m\bar{a}_c = I_C \ddot{\theta}\,\hat{k} + \frac{\ell}{2}\hat{e}_r \times m\left(\frac{\ell}{2}\ddot{\theta}\,\hat{e}_\theta - \frac{\ell}{2}\dot{\theta}^2\hat{e}_r\right) = I_C\ddot{\theta}\,\hat{k} + m\frac{\ell^2}{4}\ddot{\theta}\,\hat{k}$$

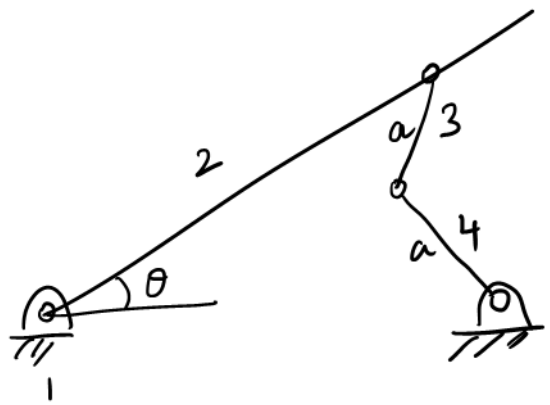
$$= \left(\frac{m\ell^2}{12} + \frac{m\ell^2}{4}\right)\ddot{\theta}\,\hat{k}$$

$$= I_O\,\ddot{\theta}\,\hat{k}$$

$$= \left[\frac{m\ell^2}{12} + m\left(\frac{\ell}{2}\right)^2\right]\ddot{\theta}\,\hat{k} = \frac{m\ell^2}{3}\ddot{\theta}\,\hat{k}$$

$$-mg\left(\frac{\ell}{2}\cos\theta\right) + V_h\left(0.8\,\ell\cos\theta\right) = \frac{m\ell^2}{3}\ddot{\theta}$$

4-bar mechanism

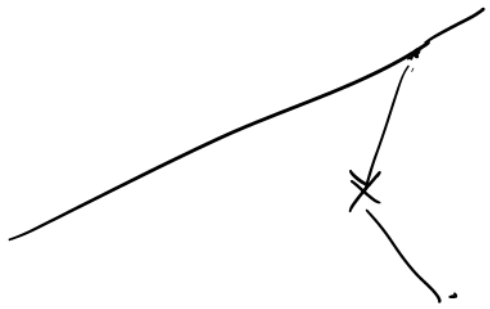


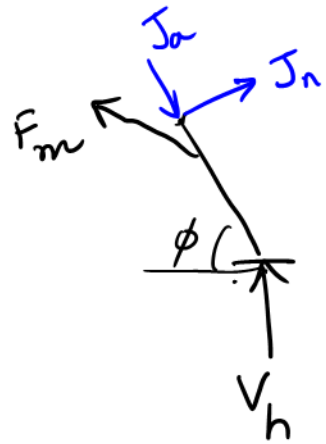
Motion capture system - time history of θ

$\theta = \theta(t)$

Force plates

→ directly measure contact forces with the ground

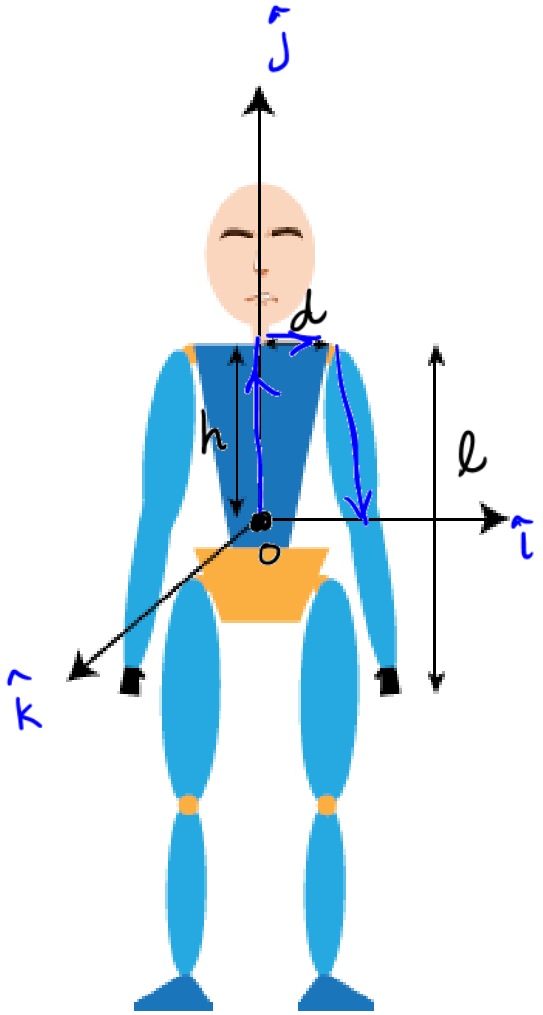


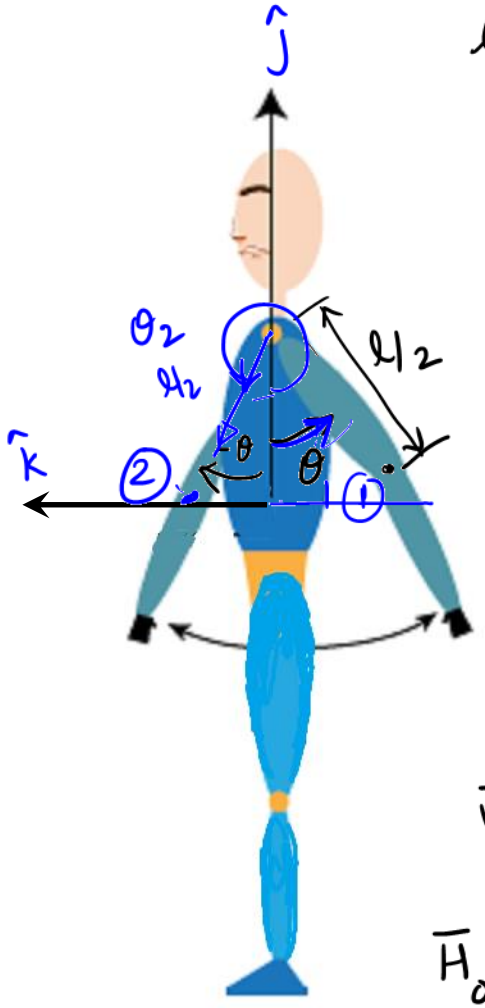


Inverse dynamics analysis

Arm swinging

Person standing straight and swinging his/her arms at a constant frequency.





$\theta_2 = 360 - \theta$

l is the length of the arm

For arm 1:

$$\bar{r}_1 = h\hat{j} + d\hat{i} - \frac{l}{2}\cos\theta\,\hat{j} - \frac{l}{2}\sin\theta\,\hat{k}$$

For arm 2:

$$\bar{r}_2 = h\hat{j} - d\hat{i} - \frac{l}{2}\cos\theta_2(-\hat{j}) + l/2\sin\theta_2(-\hat{k}) + \frac{l}{2}\cos\theta\,\hat{j} + \frac{l}{2}\sin\theta\,\hat{k}$$

$$\bar{V}_1 = \frac{l}{2}\sin\theta\cdot\dot{\theta}\,\hat{j} - \frac{l}{2}\cos\theta\cdot\dot{\theta}\,\hat{k}$$

$$\bar{V}_2 = \frac{l}{2}\sin\theta\cdot\dot{\theta}\,\hat{j} + \frac{l}{2}\cos\theta\cdot\dot{\theta}\,\hat{k}$$

$$\bar{H}_0 = \sum_{i=1}^{2}\bar{r}_i \times m_a\bar{V}_i$$

We have lumped the mass of the arm at its CoM

$$\bar{H}_{O_{arms}} = \bar{r}_1 \times m_a \bar{v}_1 + \bar{r}_2 \times m_a \bar{v}_2$$

$$= 2 d \ell_a m_a \dot{\theta}_a \cos\theta_a \, \hat{j}$$

$$\bar{H}_{O_{legs}} = 2 d \ell_\ell m_a \dot{\theta}_\ell \cos\theta_\ell \, \hat{j}$$

If you're standing and swinging your arms, the GRF must counteract the varying angular momentum.

In practice, the trunk rotates slightly to the left and right during swinging of the arms in such a way as to make $H_O = 0$ at all times.